GEK-65429

GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

MK II SPEEDTRONIC®
TURBINE CONTROL PANEL
PROTECTIVE LOGIC
TEST MODULE

GENERAL ELECTRIC

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The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, USA.

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INTRODUCTION TO MANUAL

Purpose of Manual

The purpose of this manual is to provide instructions describing the functions, operation and application of the Turbine Control Protective Logic Test Module.

The user of this manual should be an experienced electrician or technician. The user of the test module should be an operator/technician experienced and knowledgeable about the gas turbine control panel for which this test module is designed.

DESCRIPTION OF EQUIPMENT

Mechanical Information (See Figure 1)

Weight: 5 pounds (2.3 kg)

Size: Height 11 inches (27.9 cm)
Width 10 inches (25.4 cm)
Depth 5-1/2 Inches (14 cm)

Electrical Information

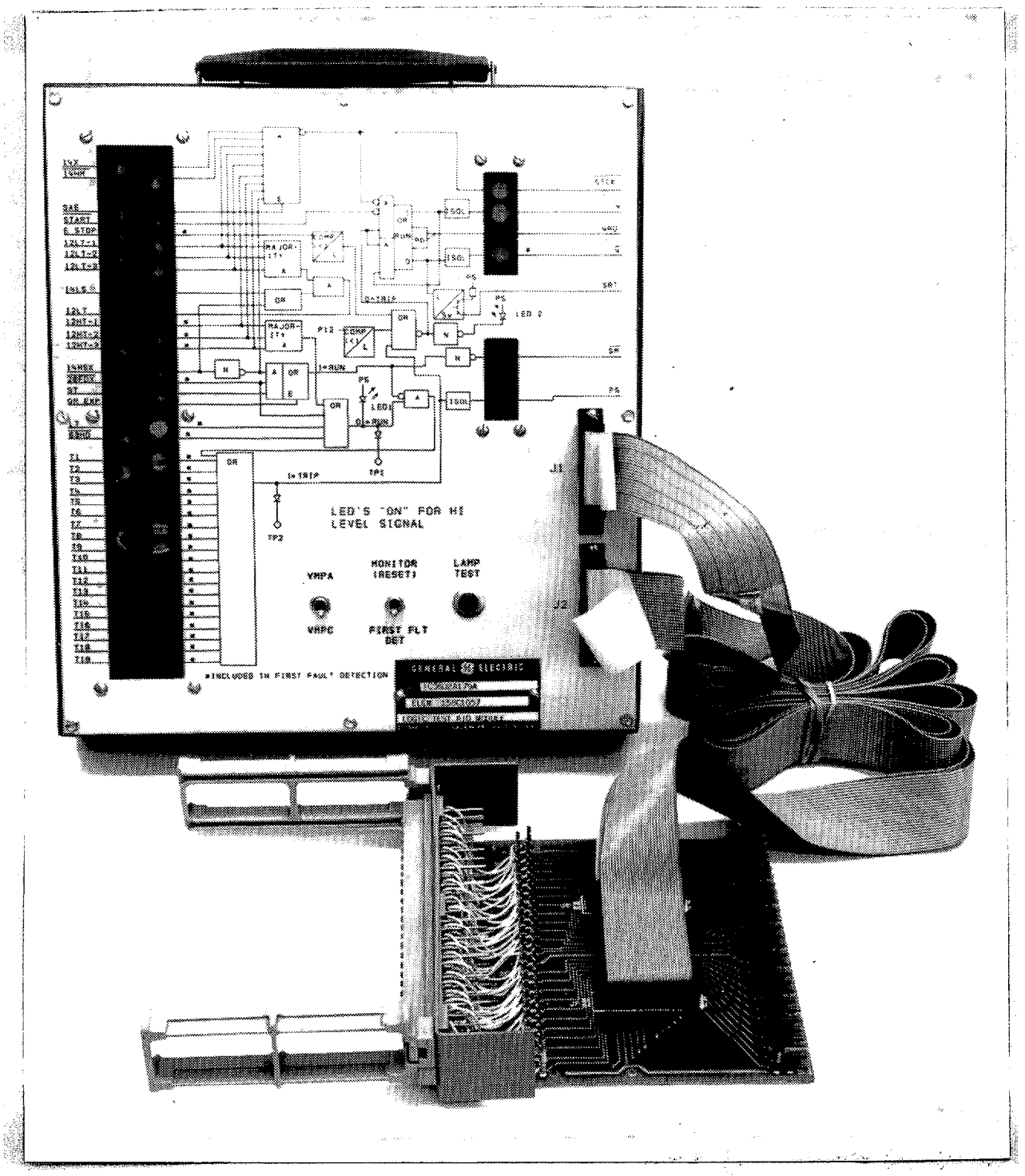
Test module power is supplied by the turbine control panel through the interface cabling, therefore no external power source is required.

Operating Environment

DIRECTO-MATIC ® Plus control equipment operates reliably in the environments typically found in heavy industry.

Standard DIRECTO-MATIC ® Plus control equipment is designed for operation in the following environment:

Temperature: 0°C to 40°C (32°F to 104°F) continuous operation, up to 50°C (122°F) for short periods. Temperature transients should be held to less than ±15°C/hour (±27°F/hour) in order to maintain optimum performance.



(DSF52704)

Figure 1
Logic Test Aid Module, Cable and Extender Board

NOTE

The maximum continuous rating of 40°C is based on providing high-reliability equipment in the long-term temperature environment realistically encountered. However, in order to ensure proper control operation under occasional high temperature conditions which might occur, an additional 10°C margin is "built-in" - i.e., the equipment is designed to operate correctly at ambient temperatures up to 50°C for short periods of time (e.g. 2-3 days). In order to maintain high reliability, extended operation (e.g. months) at this temperature is not recommended since it is generally recognized that electronic components experience a 2:1 reduction in operating life for each 10°C increase in long-term average operating temperature.

Humidity: 5% to 85% relative humidity with no condensation.

Altitude: 3500 feet (1066 meters) above sea level.

Atmosphere: Normal atmosphere found in typical control equipment locations in heavy industry.

NOTE

DIRECTO-MATIC ® Plus control equipment has been designed recognizing that the "normal" atmosphere includes various combinations of corrosive gases such as SO_2, H_2S, HC_L, etc. Based on extensive sampling techniques at numerous typical installations, General Electric Company has developed special testing procedures involving soaking for weeks at elevated temperatures in carefully selected concentrations of corrosive gases. These testing procedures simulate many years of actual installation exposure and have been used in testing all components and materials used in DIRECTO-MATIC ® Plus control equipment. This qualifying process insures the proper operation and "survival" of the equipment in actual installations.

The design has also recognized the concentrations of dust and dirt typically found in such installations. Dust made up of various iron compounds, coal, cement, paper, lint, soot, etc. has been anticipated in the design.

In addition to a careful choice of components, numerous steps have been taken in the design to further insure equipment operability - for example, all circuit runs on printed circuit boards are masked with a special plastic coating for environmental protection.

Consideration of equipment appearance and minimization of effects of adverse environments make it advisable to plan the installation to use the cleanest air readily available. For example, do not use air directly in processes which produce excessive dust and/or corrosive gas by-products (e.g. blast furnaces, slag pits, pickel lines, paper machines, etc.)

NOTE

The above specification applies to DIRECTO-MATIC ® Plus control equipment. However, some equipment, intended for special application, is designed to meet environmental requirements different than this standard. See specific equipment specifications provided with the requisition.

Front Panel

A symbol representing the VMPA/VMPC PWB mounted in a turbine control panel appears on the test module front panel. Signal presence on PWB input/output lines is indicated by LED's on the test module front panel.

A VMPA/VMPC toggle switch is set to the position corresponding to the type PWB mounted in a turbine control panel under test.

A MONITOR/FIRST FLT DET switch selects the desired operating mode. (See paragraph entitled OPERATING INSTRUCTIONS.)

A LAMP TEST pushbutton, when depressed, tests the integrity of all LED's. This switch may be used at any time that power is applied to the test module. The LED's already on will remain on steady; the remaining LED's will flash on and off, thereby conserving power while the pushbutton is depressed.

Jacks J1 and J2 are mounted on the front panel. These jacks are different sizes to prevent the cables from accidentally being plugged into the wrong jack.

Back Panel

The back panel provides storage space for the extender board and also acts as a stand which extends outward to support the test module in an upright position during use. An abbreviated set of operating instructions is included on the back panel for quick reference.

EQUIPMENT FUNCTIONS

The Protective Logic Test Module is designed for testing all MKII SPEEDTRONIC® Turbine Control Panels. The test module is used for the check-out of new turbine control panel installations by indicating the status of permissive and trip signals applied to the input of the turbine control panel master protective PWB VMPA or VMPC. The test module is also used for troubleshooting random trips on existing panels. The status of output signals from the master protective PWB may also be monitored on the test module front panel LED's.

The test module is a passive device in the sense that it is only capable of monitoring and indicating the status of signals into and out of the master protective PWB. The test module is not capable of forcing signals or inhibiting them from tripping the turbine control panel.

First Fault Detection Mode

This mode is used for either one of two conditions:

1. when the turbine experiences random trips resulting from signals which occur too rapidly for the annunciator to determine which occurred first, or

2. trip signals not directly fed to the annunciator.

This mode may be selected any time after the turbine control "4" circuit has been picked up (4 LED on). This enables first fault detection during the turbine start-up sequence and also during the "run" sequence. During the first fault detection mode, only the signal status LED's of the first trip signal into the VMPA or VMPC PWB will change states. All other trip signal indicators will be inhibited from changing state.

To determine which of two signals tripped the turbine, a 1 millisecond period (or greater) is required between the two signals. Noise filtering rejects high frequency noise which could otherwise trigger the first fault detection circuit.

Monitor Mode

The operation of the turbine trip and start-up permissive circuitry is verified by setting the MONITOR/FIRST FLT DET switch to the MONITOR position and observing the state of the LED's for proper responses. (See LED Conventions for an explanation of LED operation.)

LED Conventions

The LED's come on to indicate a high voltage level (logic 1) state and off to indicate zero or low voltage (logic 0) state.

A red LED which is on indicates that the represented signal is in the trip (high voltage) state.

A green LED which is on indicates a "run" (high voltage) condition and off for a trip (low voltage) condition.

A yellow LED is used to monitor signals whose trip status changes as a turbine transitions from start-up through a complete "run" sequence. For example, the normal condition of the overspeed signal is the high voltage state (on) during zero speed, becoming zero voltage (off) when the turbine speed exceeds that value required to cause the 14HS signal to become logic 1. Typically, for speeds greater than 14HS, yellow LED's come on indicating a tripped condition.

OPERATING INSTRUCTIONS

Module Connections

Prior to connecting the test module to the turbine under test, ensure turbine control panel power is off.

Remove the PWB (VMPA or VMPC) to be tested from the turbine control panel.

Insert test module extender board, with attached cables into the slot from which the PWB was removed.

Insert the removed PWB (VMPA or VMPC) into the extender board.

Connect free end of both test cables to J1 and J2 connectors on test module.

Switch Settings

Set the VMPA/VMPC switch to the position corresponding to the type PWB inserted into the extender board.

Apply power to the turbine control panel.

The test module is now ready for use.

Monitor Mode

Depress LAMP TEST pushbutton and observe that all LED's come on. Set MONITOR/FIRST FLT DET switch to MONITOR position and observe the status of the LED's. (See paragraph entitled LED Conventions.)

First Fault Detection Mode

Depress LAMP TEST pushbutton and observe that all LED's come on.

Set the MONITOR/FIRST FLT DET switch to the FIRST FLT DET position and observe the status of the LED's. All LED inputs applicable to the FIRST FLT DET mode are marked with an asterisk on the module front panel. These LED's should all be in the RUN state.

NOTE

The turbine control panel "4" circuit must be picked up before selecting the first fault detection mode.

The test module is now set to capture the first signal to trip the protective system of the turbine control panel.

MODULE MAINTENANCE INSTRUCTIONS

Periodic Maintenance and Inspection

There are no periodic maintenance requirements for this test module.

Prior to each use of the test module, inspect the module, extender board and both ribbon cables for any obvious defects.

Self Test

The module contains one PWB with plug-in LED's mounted on it. The LED's and LED drivers are checked by depressing the LAMP TEST pushbutton while observing that all LED's come on.

Troubleshooting

If a LED fails to come on, replace it with a new LED or exchange it with a known good LED mounted on the PWB. If this does not correct the problem, remove and replace the DS3800HTAA PWB.

The electronic circuits and cables can be checked with the test module connected to the turbine control panel and with the turbine shut down. Use standard DIRECTO-MATIC ® II turbine operating procedures to manipulate the turbine control panel signals and observe that the associated test panel LED(s) tracks the signal.

An interconnection diagram (Figure 2) is provided for continuity checks and signal tracing in the event that the interface cabling is suspected.

The first fault detection circuit may be tested with the test module connected to a turbine control panel with the FIRST FLT DET mode selected. During a normal turbine shutdown, observe that only one LED within the first fault detection circuit (indicated by asterisks on front panel) indicates a trip condition. The signal which trips the turbine varies from one panel to another depending upon whether it is a utility drive, pumper, etc.

If the first fault function does not operate properly, but the monitor mode does, the HTAA PWB should be suspected. However, check the toggle switch and associated wiring before replacing the PWB.

Disassembly Instructions

To gain access to the DS3800HTAA PWB, remove eight screws at the edge of the front panel, then lift front panel free of the module housing.

Part numbers of field replaceable parts are listed in the last paragraph of this publication.

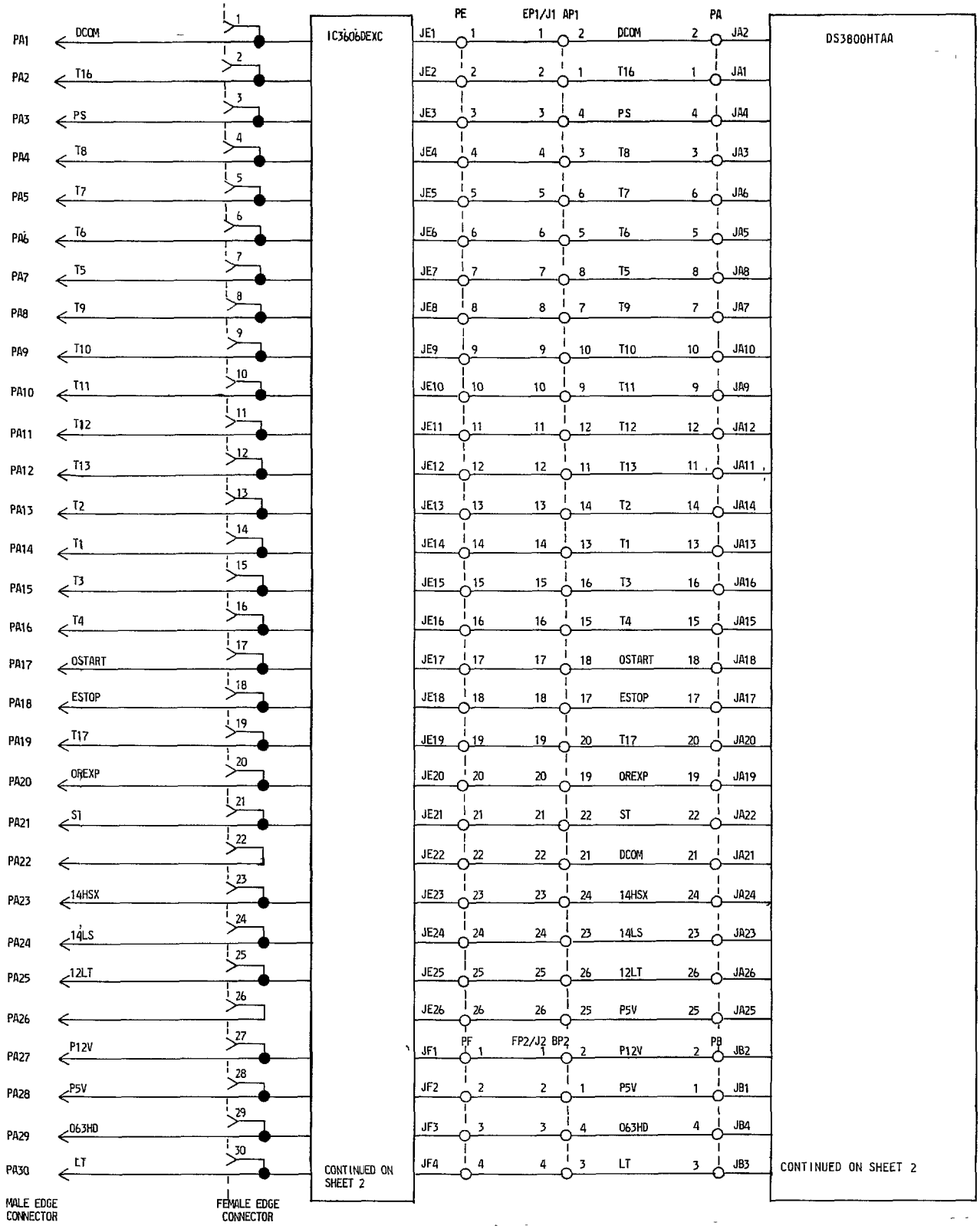


Figure 2
Protective Logic Test Aid Module Interconnection Diagram
(Sheet 1, G.E. Drawing 158C1057)

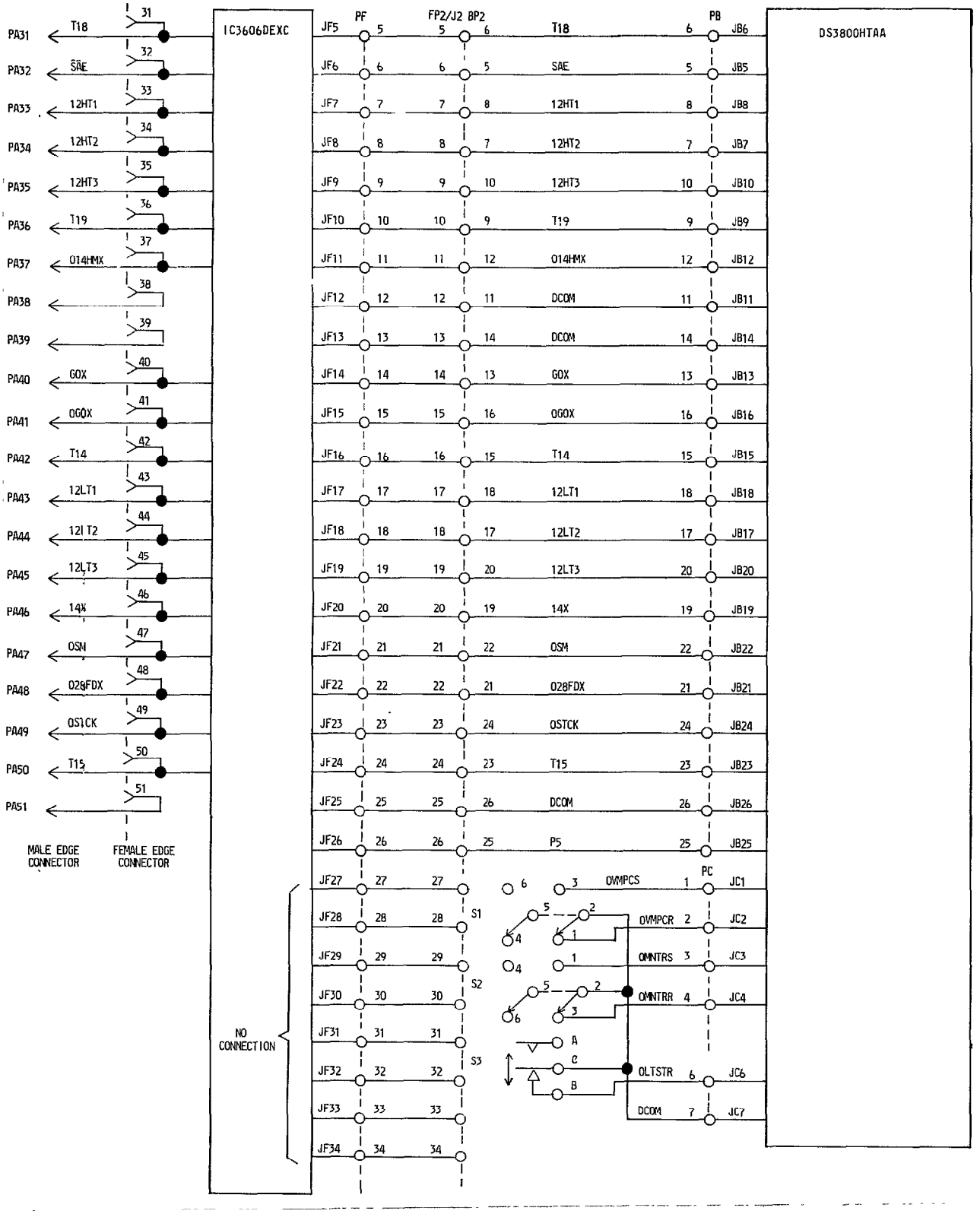


Figure 2
Protective Logic Test Aid Module Interconnection Diagram
(Sheet 2, G.E. Drawing 158C1057)

FIELD REPLACEMENT PARTS

Following is a list of field replaceable parts:

Nomenclature	G.E Catalog No.
LED, Yellow	218A4867P1YL
LED, Red	218A4867P1RD
LED, Green	218A4867P1GN
PWB	DS3800HTAA1A
Connector, J1	218A4547P3
Connector, J2	218A4547P4
Extender Board	277A6557G1
Ribbon Cable (34)	DS3815RCG1AL072
Ribbon Cable (26)	DS3815RCE1AL072
Ribbon Cable (26)	DS3815RCE1CL005
Pushbutton	68A7332P6
Toggle Switch	218A4861P1

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